



Question Set

Question 1

A one-day loss L has values 0, 2, 7, and 15 million with probabilities 0.75, 0.15, 0.08, and 0.02. Compute $E[L]$, $\text{Var}(L)$, and the 98 percent VaR under the loss convention.

Question 2

Prove the inclusion-exclusion identity for two events and apply it to

$$\mathbb{P}(A) = 0.4, \quad \mathbb{P}(B) = 0.35, \quad \mathbb{P}(A \cap B) = 0.12$$

Question 3

A rare jump loss has probability p and size J . Show why it can dominate expected loss even when p is small.

Question 4

State the axioms of probability and explain which axiom rules out negative scenario probabilities.

Question 5

Define VaR as a quantile of a loss distribution and discuss what happens when the CDF has jumps.

Question 6

Let L be a nonnegative loss. Prove Markov's inequality and explain its risk interpretation.

Question 7

Two loss events are independent with probabilities 3 percent and 5 percent. Compute the probability of at least one event, and compare with the simple sum.

Question 8

Explain the difference between real-world probability and risk-neutral probability in one paragraph.

Question 9

A risk analyst estimates $P(L > 10) = 0.01$ from 250 observations. Why is this estimate statistically fragile?

Question 10

Define a sigma-algebra and explain why it appears in conditional risk measurement.

Question 11

If

X

is integrable, prove

$$E[X] = \int_0^{\infty} \mathbb{P}(X > x) dx - \int_0^{\infty} \mathbb{P}(X < -x) dx$$

Question 12

Why is expected shortfall considered more informative than VaR for tail risk?